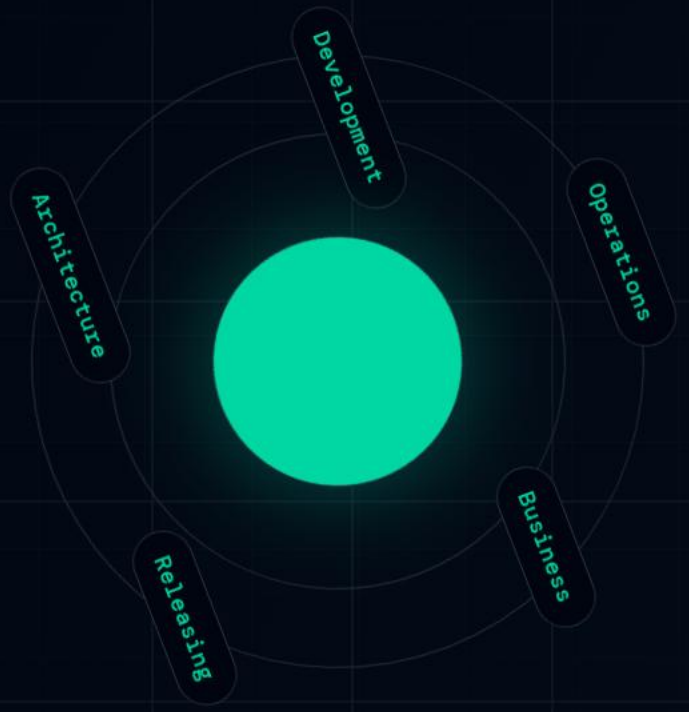
AI-NATIVE ENGINEERING SYSTEMS

Beyond *Copilots.*

Context-driven workflows for modern R&D teams

PRANATH PATHAK



Hello, I'm *Pranath Pathak.*

DevOps & Platform Engineer

Nokia — Autonomous Networks

linkedin.com/in/pranathpathak

pranathpathak6@gmail.com

SCAN TO CONNECT

THE REAL BOTTLENECK

The code was never *the three-week part.*

DAYS

Write the change

Once the requirement was clear.

WEEKS

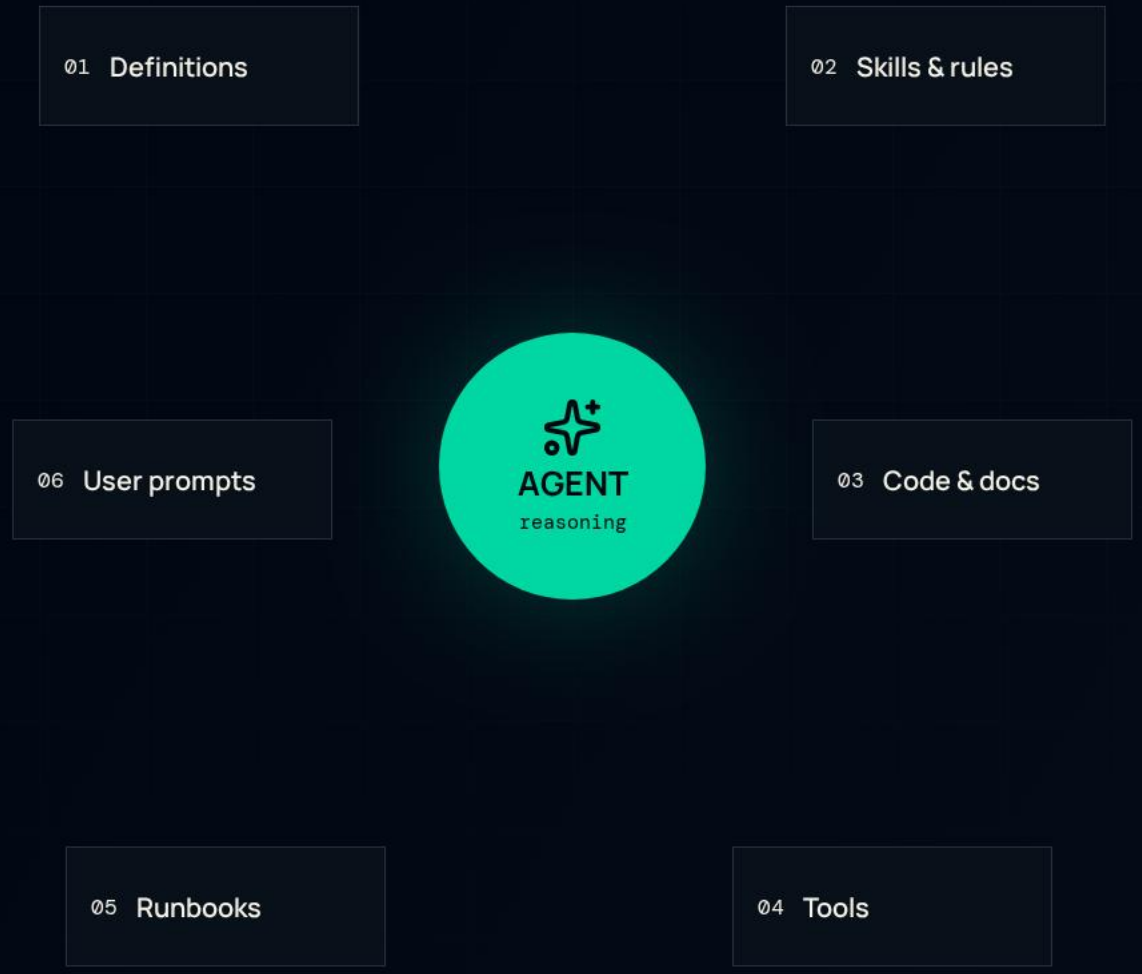
Wait through handoffs

Requirement → design → plan → build → test → operate

Every handoff needed a meeting because the decision lived in *someone's memory.*

THE PREMISE

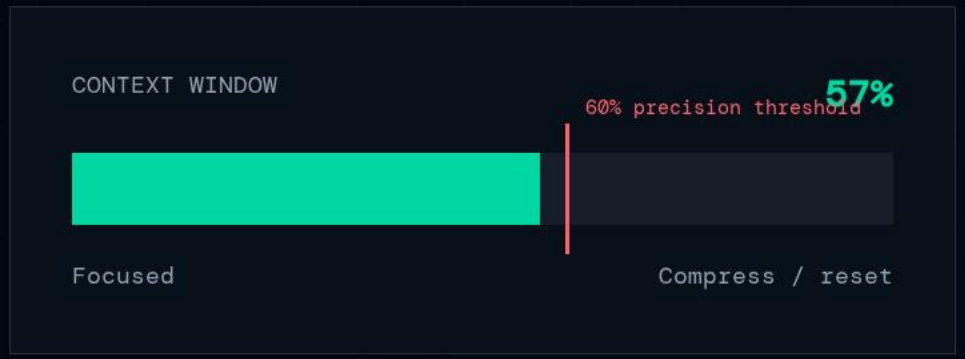
The model is not the system.



Performance comes from supplying the *right context* at the right moment.

CONTEXT IS A BUDGET

Protect the working memory.



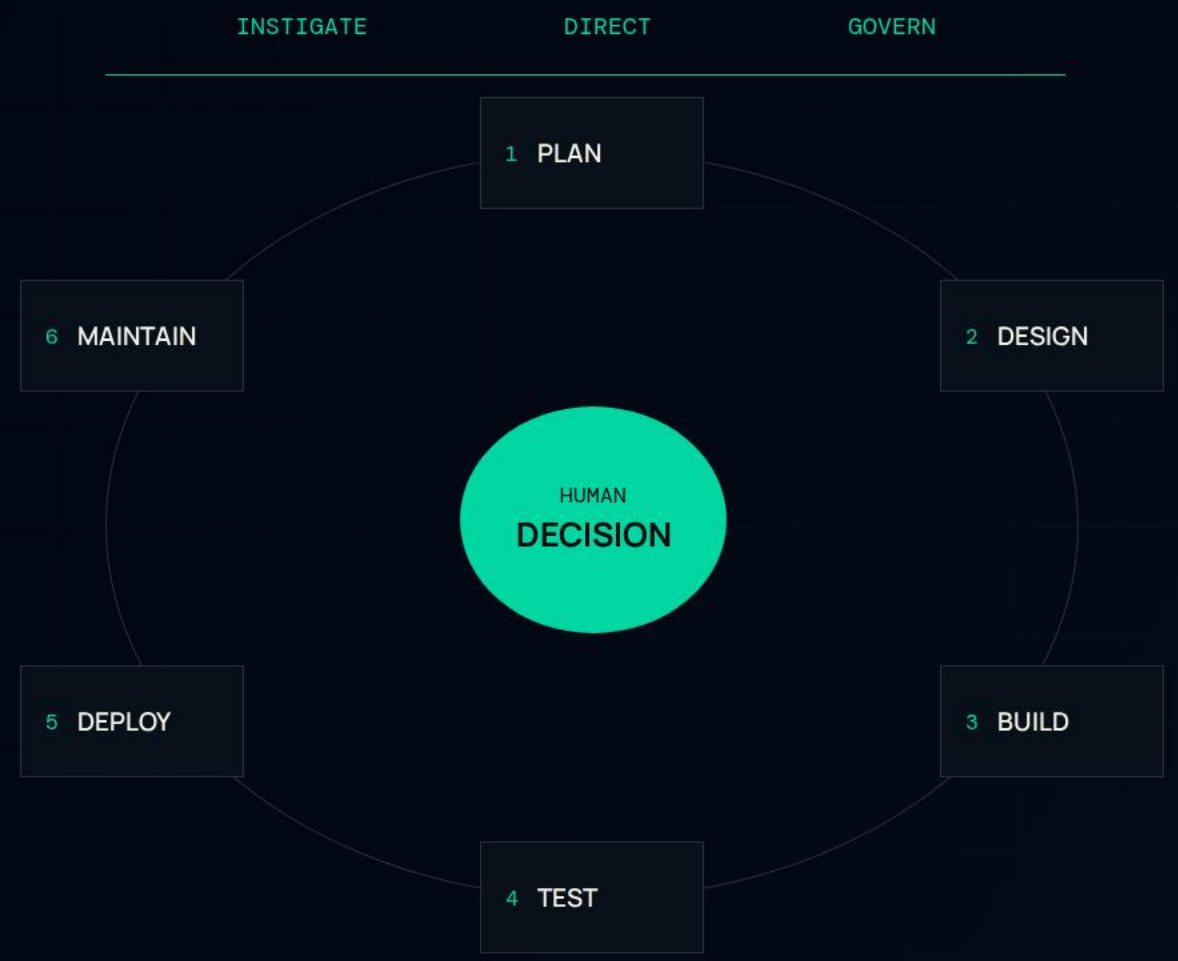
<60%

Keep the live session below the saturation threshold.

1 Load only the node → 2 Complete one bounded task → 3 Write decisions back → 4 Reset cleanly

AI-NATIVE SDLC

A fast pen. *No authority.*



The agent drafts, investigates and implements. *A named person commits.*

CONTEXT STORES / NODES

One Git repo per decision domain.

BUS — Business Store

- PLM intent
- Feature epics
- Non-goals

ARC — Solution Architecture

- Architecture runway
- Studies & ADRs
- Enabler epics

DEV — System Design & Dev

- Domain harnesses
- Pod organization
- Code & stories

OPS — Incident Response

- Execution workspace
- Diagnosis & RCA
- Sanitized evidence

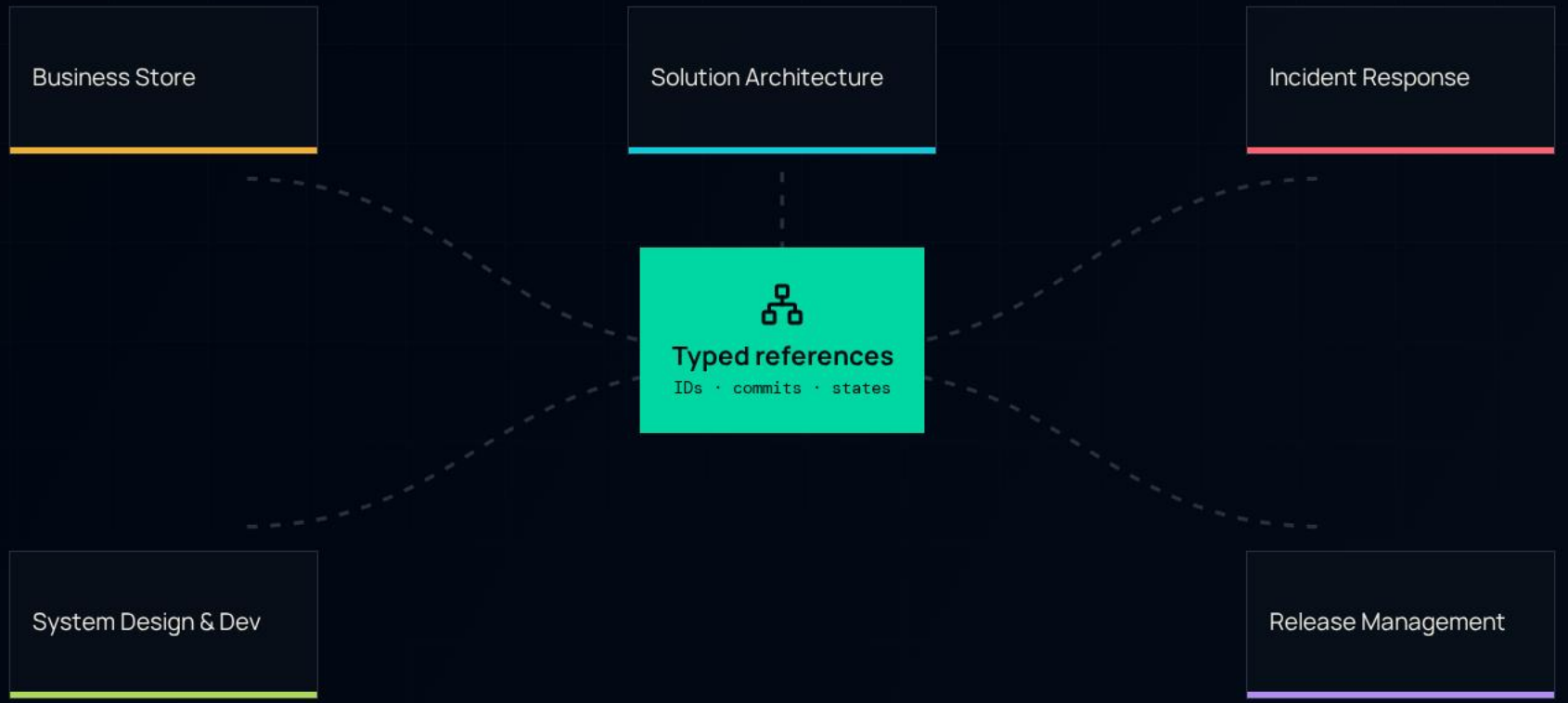
REL — Release Management

- Commitments
- Release lines
- Readiness checks

Versioned · reviewable · domain-owned · agent-readable

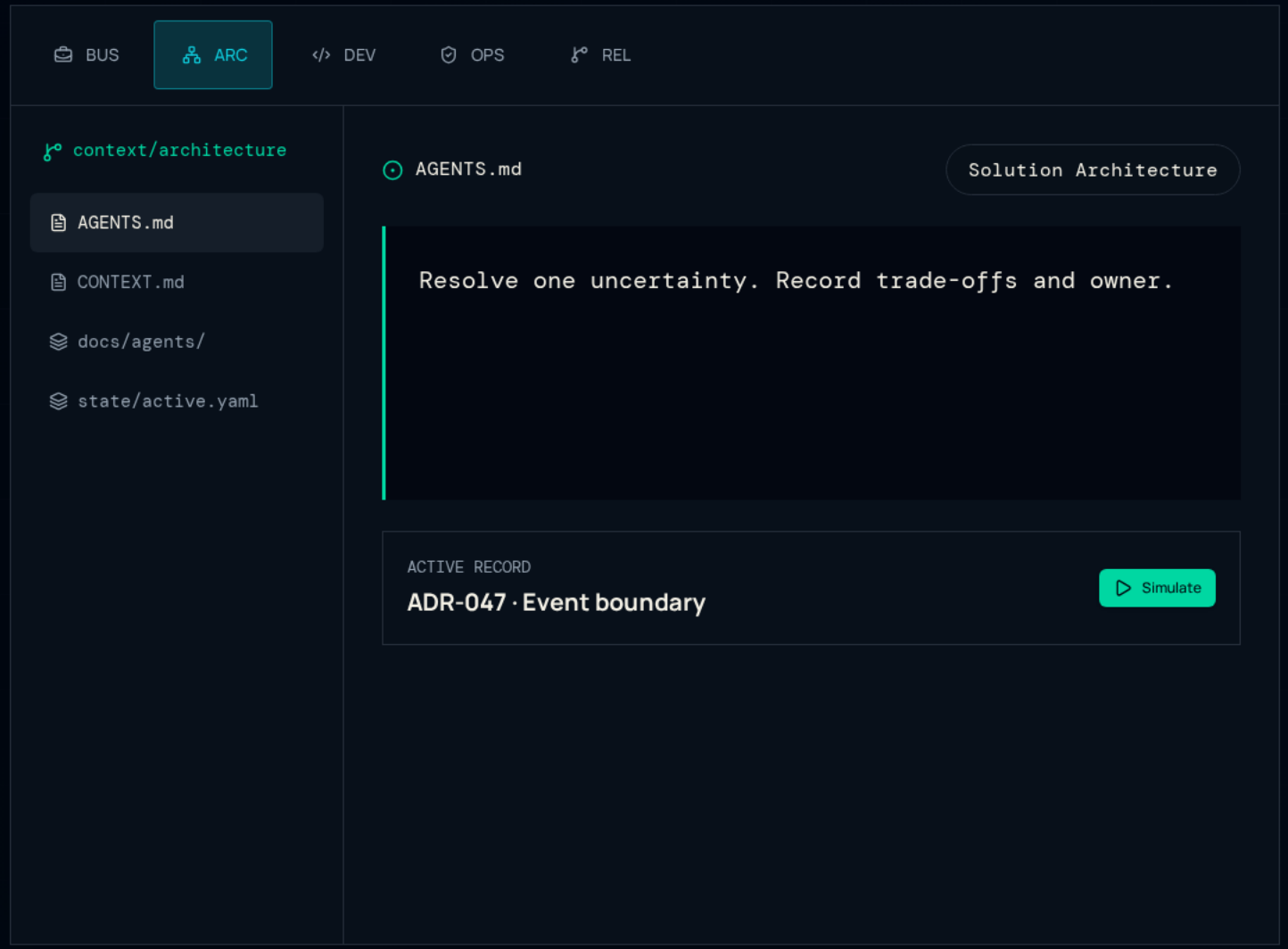
THE INTEGRATION LAYER

The graph is the product.



No central database. *Coherence emerges from references.*

Open the store that owns the decision.



FROM BUSINESS ASK TO BUILDABLE WORK

Architecture turns intent into *no-guessing work.*

01	02	03	04	05	06
Request	**Focused study**	**Feature increment**	**Domain enablers**	**Acceptance**	**Ready**
Outcome + boundaries	One hard uncertainty	Releasable customer slice	One team per package	Product + domain evidence	Human decision

DDD
Design the seams and ownership.

BDD
Write acceptance as part of the specification.

RUNWAY
Optimize the whole solution, not one epic.

THE OPERATING MODEL

Four domains. *One living record*

01

SOLUTION

Intent · runway · architecture · acceptance

02

IMPLEMENTATION

Stories · code · tests · runbooks

03

RELEASE

Commitment · evidence · publication

OPERATIONS RUNS THE WHOLE TIME

Observe → diagnose → return work to the domain that must act

An item moves when it is ready. *The rest of the backlog does not wait.*

THE QUALITY BOUNDARY

Ready means *no guessing.*

READY FOR AN AGENT

A person could leave, and the agent would not have to invent the requirement.

- ✓ The change is named
- ✓ Limits are explicit
- ✓ Proof of done is written
- ✓ Decisions are linked

← QUESTION FOUND Send the work back. That return is a success of the process.

ADOPT WITHOUT A BIG-BANG REORGANIZATION

Phase 1.
Turn the stores on.

ENTRY A

Business request

Enter the architecture runway. Shape a solution and domain breakdown.

ENTRY B

Operational signal

Diagnose the issue. Route actionable work into the same implementation flow.

KEEP ✓ Current Scrum teams ✓ Current code structure ✓ Current release process

SCALE THROUGH DOMAIN OWNERSHIP

Phase 2.
Organize for flow.

ARCHITECTURE RUNWAY

Feature → domain enablers

POD 1

DOMAIN A

Design · code · test · operate

POD 2

DOMAIN B

Design · code · test · operate

POD 3

DOMAIN C

Design · code · test · operate

CONTINUOUS VALIDATION — INTEGRATE — PARALLEL RELEASE LINES

Decoupled domains let plans be agreed and delivered *incrementally.*

FOUR GATES · FOUR OWNERS

Release reads. *It does not invent.*

01	02	03	04
Solution-ready	**Implementation-ready**	**Engineering-done**	**Release-ready**
Architecture	Product + Engineering	Development + QA	Release management

RELEASE LINE — Internal artifacts · External artifacts · Readiness checklist

THE DAILY OPERATING RHYTHM

Read. Decide. *Write it back.*

01

READ

Open the store that owns the decision.

02

DECIDE

Complete one bounded task with human review.

03

WRITE BACK

Commit the learning where the next session finds it.

START MONDAY

Choose one workflow. Name its owner. Create its first record.

HUMAN-GOVERNED

THANK YOU

Questions?

Code is no longer the bottleneck.
The record is the product.
People still decide.

PRANATH PATHAK linkedin.com/in/pranathpathak